SOURCE-SPECIFIC THEOREM REVIEW

ATLAS: Re-Audit After Repair

Two Armies of One: Certified Scout–Probe Rendezvous in Proof Search

Version 0.3 · 26 October 2026 · published predecessor v0.2

Disposition: PAPER-LEVEL PASS, UNDER THE EXPLICIT HYPOTHESES. The two specification defects identified in the v0.2 audit are corrected in v0.3. Its formal mathematical statements are defensible as conditional theorems in a fully checked, MP-only certificate grammar. The protocol is not proved faster, a useful early encounter is not guaranteed, and no full proof-assistant formalization or institutional external review is claimed.

Materials checked. The complete v0.3 LaTeX source, its compiled nine-page PDF, comparison with v0.2 and the 26 October independent-style audit report, and a finite illustrative test program containing positive and negative certificate controls. Re-audit checks the new text, not only the original repair ledger. This report uses "ATLAS" as the requested audit role; it is assistant-performed review.

1. What was repaired and why

1. **Verifier closure (material).** Original v0.2 stipulated sound terminating checking but did not require an otherwise valid graft to be accepted. New Definition 2.1 defines correct finite MP proof DAGs and an *iff* checker, with explicit subproof heredity, opening of a cut occurrence, and substitution/graft closure. Thus Theorem 3.1 has the hypothesis its proof needs.

2. **Proof-specific compatibility (material).** Original Theorem 4.2 identified $H \vdash L$ with eventual fixed-Scout meeting without quantifying over acceptable Probe receipts. Repaired Theorem 4.2 first gives the exact criterion $\exists \pi \in E_P(L)\, \mathsf{Comp}(\pi, \rho_L)$; the simpler $H \vdash L$ iff is stated only with formula completeness *and uniform compatibility*. Certificate completeness supports a separate existence-of-compatible-proof formulation.

3. **DAG opening and several obligations.** Proposition 6.3 expressly relies on hereditary extraction and opening; Proposition 8.2 requires jointly compatible finite discharge receipts and suitable Probe fairness. This prevents importing an arbitrary accepted certificate into an incompatible proof scope.

4. **Chronology and provenance.** The first-settlement time is $s_T = +\infty$ if the unassisted branch never settles. The manuscript names its checks as internal paper-level review and uses the requested author name form `<Brian Tenneson>` in both typeset credit and PDF metadata.

2. Theorem-by-theorem disposition

Result	Audit	Reason
Thm. 3.1 / proof graft	PASS	With the explicit full local MP grammar, fresh-copy insertion preserves the DAG order, authorized leaves and ordered MP ports; the checker accepts the resulting valid proof.
Return-decoder corollary	PASS	Explicitly conditional on a separately validated translation and original checker. No implicit cross-calculus proof is claimed.
Thm. 4.2 / fixed Scout	PASS	The proof-relative $E_P(L)$ criterion is exact under continuing fair matching; the corollary to $H \vdash L$ has the required uniform compatibility and formula-coverage hypotheses.
Thm. 5.3 / route-changing Scout	PASS	A chosen finite compatible witness is eventually recorded and tested under exhaustive certificate enumeration and persistent, continuing fair search.
Prop. 6.2 / Mid discovery	PASS	The finite syntactic filter is terminating; this is a direct instance of Theorem 5.3.
Prop. 6.3 / internal DAG cut	PASS	Extraction and opening are declared checker-closed operations; the specified ancestor and descendant conditions exactly give a target-free proper witness.
Prop. 8.2 / finite discharge	PASS	Finite repeated grafting works for jointly compatible certificates; certificate-complete search finds the chosen witnesses.
Antichain readiness	PASS	Inclusion-minimal supports preserve the disjunctive readiness predicate for a fixed family only, with shared-context compatibility.

3. Adversarial counterchecks

Bounded verifier obstruction. For $H = \{p, p \to \ell, \ell \to t\}$, a sound verifier capped at one MP step accepts separately the prefix $H \vdash \ell$ and suffix $H, \ell \vdash t$ but rejects the two-MP splice. The new checker hypothesis explicitly rules this out. It is a countermodel to the *old* weak verifier assumptions, not to new Theorem 3.1.

Stranded Scout. The valuation $p = t = \text{true}$, $q = \text{false}$ satisfies $\{p, p \to t, q \to t\}$ and falsifies q; the fixed Scout's route through q cannot be discharged despite $H \vdash t$.

Three midpoint controls. A target-first derived $L = t \wedge (q \to q)$ is rejected by the target-free proof ancestry test; a genuine $p \to \ell \to t$ cut can be found after independent settlement; and a closed $q \to q$ bridge from $H = \{t\}$ can be structurally legal yet logically inessential. All three examples support *limited* claims and do not imply an advantage.

Finite illustrative execution. A small standalone program checked 10 cases: accepted one-step forward and conditional proofs, accepted their two-step graft with a full local checker, reproduced rejection of the graft by the sound one-MP-cap checker, and rejected swapped MP ports, incorrect conclusions, unresolved open assumptions, and backward/cyclic dependency indices. Those tests only illustrate the finite rules; they do not machine-verify arbitrary quantified theorems.

4. Remaining scope boundaries

1. A proof of T yields at best the degenerate cut $L = T$ without more assumptions. Existence of a *genuine* target-free intermediate proof occurrence needs the additional witness required in Proposition 6.3.
2. Fairness gives eventual encounter under unbounded execution, not an effective universal time bound or a proof that first rendezvous beats direct settlement.
3. Mid is decidable for a supplied finite proof triple. Deciding whether *some* eligible triple exists for a formula is not established.
4. Translation from an MP-only presentation back to a different first-order calculus depends on the external decoder's correctness, not established here.
5. Real AMLD instances, machine-checker integration, charged runtime benchmarking, and empirical speedup remain unperformed.

5. Release gate and disposition

Mathematical paper-level release gate: PASS for version 0.3 as a preliminary, internally re-audited research note. This means the known overclaims are corrected and the conditional results have arguments matching their hypotheses. It does *not* mean independently certified mathematics, novelty certification, completeness of all conceivable audits, or proof-assistant checking. Retain the site label "ATLAS-REPAIRED PRELIMINARY RELEASE" rather than "FORMALLY CERTIFIED." Preserve published v0.2 as a historical predecessor and expose v0.3 as the current corrected edition.